UNIVERSITY OF WATERLOO

SYDE 556/750 – Simulating Neurobiological Systems

Practice Test 3C: Nengo & Simple Dynamics

Student ID: _____ Time: 30 minutes Total: 20 marks (plus 2 bonus)

- **Closed book.** No notes, textbook, calculator, phone, or other devices.
- The marks for each question are shown in brackets on the right.
- Write your answers in the space provided. Show your work — partial credit is given for correct reasoning.
- Unless stated otherwise, every synapse is the first-order low-pass $h(t) = \frac{1}{\tau} e^{-t/\tau}$.

Question 1. **[4 marks]**

(a) A 1-D LIF neuron has a positive encoder. Its normalized intercept is 0.25, and its maximum firing rate is 100 Hz when the represented value sits at the edge of the radius. On the left, sketch its tuning curve for `radius=1` and for `radius=2`, on the same axes, for $-2 \leq x \leq 2$. Label where each curve starts firing and where it reaches 100 Hz.

(b) A neuron in a 2-D ensemble has `radius=3`, encoder $e = (-1, 0)$, normalized intercept $1/3$, and maximum rate 90 Hz at the radius along its encoder. On the right, sketch its firing rate along the line $\mathbf{x} = (s, 0)$ for $s \in [-3, 3]$. Label where it starts firing.

(a)	(b)

Question 2. **[4 marks]**

Both connections of a population use the synapse $\tau = 0.1$ s.

(a) Choose the transforms so the population implements the leaky integrator

$$\dot{x} = -\frac{1}{5}\,x + v.$$

With $v = 0$ and $x(0) = 0.4$, write $x(t)$ and give its value at $t = 5$ s.

(b) Using the same τ, choose the transforms for

$$\dot{x} = -20x + 20u.$$

Sketch the response to a unit step in u from $x(0) = 0$. On the same axes, sketch the response of the synapse alone (a low-pass of time constant 0.1 s, with no recurrence). Which curve settles faster?

Question 3. **[4 marks]**

A population with radius 2 and synaptic time constant τ has no input, $g(u) = 0$, and recurrent connection $f(x) = x^3$.

(a) Write $\tau\dot{x} = \varphi(x)$. Find every equilibrium and say whether it is stable or unstable.

(b) Sketch $\varphi(x)$ for $-2 \le x \le 2$ with arrows on the x-axis. Describe the motion from $x(0) = 0.5$ and from $x(0) = 1.2$.

Question 4. **[4 marks]**

A 2-D population implements the oscillator

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \mathbf{x}, \qquad \omega = \pi \text{ rad/s},$$

with recurrent synapse $\tau = 0.05$ s.

(a) Give the recurrent transform A'. Leave π in your answer.

(b) After a short kick the state is $\mathbf{x}(0) = (0.6,\ 0.8)$. Write $x_1(t)$ and $x_2(t)$ in terms of $\sin(\pi t)$ and $\cos(\pi t)$. In the box, sketch both signals for $0 \le t \le 2$ s, and sketch the trajectory in the (x_1, x_2) plane with an arrow for its direction.

Question 5. **[4 marks]**

An integrator $\dot{x} = u$ is built with recurrent transform 1, recurrent synapse $\tau = 0.1$ s, and input transform τ. It starts at $x(0) = 0$, and the ensemble radius is 2. The input is the pulse

$$u(t) = \begin{cases} 3, & 0 \le t < 0.4 \text{ s}, \\ 0, & t \ge 0.4 \text{ s}. \end{cases}$$

(a) Find the ideal $x(t)$ and sketch it.

(b) The input connection uses a faster synapse, $\tau_{\text{in}} = 20$ ms. Sketch the decoded $x(t)$ on top of your answer to (a). How large is the gap just before $t = 0.4$ s, and what value does x settle to after the pulse?

Bonus Question 1. **[2 bonus marks]**

The integrator of Question 5 is rebuilt with recurrent transform 0.98 instead of 1, and with matched synapses. There is no input, and it is holding $x(0) = 1$. What dynamical system is it actually implementing? How long until the held value has fallen to 0.5? Sketch $x(t)$.